

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Extracted features from quantitative and textual data, writing a *Python* algorithm that pulls evidence segments from debate cases, validates each source URL and removes statistical outliers via **IQR** on word-count distributions
- Applied rigorous data quality checks, catching **30%** of stored evidence as duplicates and shrinking index bloat
- Worked in an *Agile* team of **10 engineers** to build the *Python* (*FastAPI*) search engine behind a debate research platform with **10K+ visitors**, indexing **600K+ documents** on *Elasticsearch* with results in **under 1 second**
- Ranked search results by recency so evidence on the current month's debate topic surfaces first for researchers

Product Development Intern | Jetson

January 2025 – February 2025

- Designed an **A/B test** on 30 test users to validate a redesigned *Figma* onboarding flow that cut **12 pages to 5**
- Ran **7 iterations** of UI/UX experiments on 3 features of a **100K+ user** app, analyzing results after each round
- Communicated data-driven recommendations to optimize user engagement to the CEO in a **5-page report**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Cleaned, normalized and analyzed 3 years of Hootsuite analytics data (**200+ TikTok posts**) in *Python* using open-source libraries such as *pandas*, then cross-checked engagement trends in *Power BI* and *Microsoft Excel*
- Found the key insight: "science fun facts" posts drove 69.3% of views while making up only 7% of all content
- Presented the insight to the Executive Director; the resulting strategy lifted engagement **24% (8K+ views)**

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Mined **15K** truck-listing images, auditing quality by dropping duplicates and non-USD entries to normalize prices
- Extracted condition, brand, model and year features from photos via a *JSON schema* for **Claude Sonnet 4.6**
- Validated the extracted features with exceptional attention to detail, revising the *JSON schema* after every misread label until **10 images in a row** came back correct on all 3 features, fine-tuned heavily before modeling
- Fit a *quantile regression* model to those features to predict truck market prices at **87% accuracy** over **30 samples**
- Headed a **4-person** team building Kamion's (**YC S22**) pricing tool. **2nd Place at UIUC Founder's Hackathon**

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Conducted an independent data project on waste classification, publishing the full research methodology as an *arXiv* paper and releasing 2 original image datasets on *Kaggle*. **Placed 2nd at the Arizona State Science Fair**
- Trained a two-stage neural network (*EfficientNetV2-S*, **4.39M** parameters) in *Python* using the open-source *PyTorch* library on **15K+ images**, reaching **96.13% accuracy** when routing waste into 4 distinct disposal categories
- Engineered a synthetic data pipeline (*U2-Net*, alpha-compositing) generating **9K** contamination training images
- Integrated the model with an *Arduino Uno* smart bin prototype that sorts each waste item in **under 2 seconds**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Normalized and validated vendor data from 2 APIs (*OpenAlex*, *Semantic Scholar*), backfilling missing metadata
- Loaded records into *PostgreSQL* (*Supabase*), writing *SQL* queries to slice papers by field, year and citations
- Visualized the outbound citations of each paper across **26 research fields** in an interactive *React Flow* graph
- Deployed a *Claude Sonnet 4.5* **RAG** agent that extracts useful information from per-session text embeddings

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Data & Analytics: pandas, NumPy, Microsoft Excel, Power BI, A/B Testing, Hootsuite

Concepts/Practices: Data Validation, Data Quality, Data Cleaning, Statistical Analysis, Data Mining, ETL, Agile

AI/ML: scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, Hugging Face, Claude API, Grad-CAM, Gemma 4

DevTools: Kaggle, Git, GitHub, Docker, VS Code, Claude Code, Codex, CI/CD, Figma, Bun, Arduino IDE